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Discussion of the Feasibility of the Vacancy Chain

December 9, 2013

Summary

Rousseau once said Man was born free, and he is everywhere in chains. However, it is not trying to spread a pessimistic world outlook. Instead, it is through interaction among people that our society can make progress.

Just like hermit crabs would group together to exchange shells so that every individual can get a larger size of home. Our human society also exists similar phenomenon which is called vacancy chain. However, allocating shells among is relatively a easy job, how to exchange job among people is a rather difficult problem.

In order to address problems above. We conclude three sub-problems and its solution in our paper: 1) Figure out the conditions need to meet so that vacancy chain can function as a tool to optimize every individual can benefit. 2) Build a model to evaluate ones competencies and propose an algorithm as social strategies to exchange jobs in vacancy chains so that everyone benefits. 3) Develop a inequality to answer how many job applicants do need to satisfy both employers and employees and run our model to predict the change of HR expends for some enterprise customers within the next 5 years.

In the first sub-problem, we discuss the different effects of different organizational hierarchies on vacancy chains. Then we successfully figure out that hierarchical organizational forms with less middle managers than straight pyramis are most likely to be regulated by vacancy chains. And fortunately, most enterprises' organizational forms are similar to it. That is to say, vacancy chain can be applicable to most cases. Next, in second sub-problemswe propose a linear equation as a evaluating function and get a overall score for every individual which represents one's competencies. Then, based on the overall scores, we use a greedy algorithm to exchange jobs among people in vacancy chain. Peoples qualities will be compatible with the job requirement. In this case every individual will get the most appropriate job. Finally, in the third sub-problemsince vacancy chain also has its cost like replacement cost and training cost. We will not allow the total cost outweigh its benefits. So we will propose a inequation that need to meet so that the cost will not outweigh the benefits. That is to

say, we will figure out how many job applicants do need to satisfy both employees and employers at certain situations. Finally, we will run our model to show the change of HR expends for some enterprises customers within the next five years.

In the whole modeling process, we give full consideration to validity, feasibility and cost-efficiency of our model.

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1 Introduction

As we know, the concept of the term-vacancy chain is derived from the scientific observation from a kind of phenomena in hermit crab population. As the hermit crab grows in size, it has to find a larger shell and abandon the previous one. Several hermit crab species, both terrestrial and marine, use vacancy chains to find new shells: when a new, bigger shell becomes available, hermit crabs gather around it and form a kind of queue from largest to smallest. When the largest crab moves into the new shell, the second biggest crab moves into the newly vacated shell, thereby making its previous shell available to the third crab, and so on. Since this biological discovery, more meanings have been added to the term of vacancy chain.

Nowadays, a vacancy chain is a social structure through which resources are distributed to consumers. In a vacancy chain, a new resource unit that arrives into a population is taken by the first individual in line, who then leaves his or her old unit behind, this old unit is taken by a second individual, leaving his or her old unit behind, and so forth. Usually when a vacancy occurs the bulk of the people who fill the positions come from the inside the internal labor market rather than hiring someone new. The individuals being moved and or promoted to a new job or position often gain a higher salary of more benefits which has an important impact on the organization.

In this paper, we propose a model of vacancy chain to maximize the total interests of employers and employees, then use it to predict the change of job movement in some enterprises. We conclude three sub-problems to tackle in our paper.

We discuss when the vacancy chain can be used as a tool to optimize the jobs allocation among employees and what conditions need to meet so that vacancy chain can exert its function.

Build a model to calculating employee's competencies which can be used to arrange employees when vacancy chain happens and develop an algorithm of job allocation which tries to benefit every employee.

Discuss how many job applicants do need to satisfy both employers and employees and Run our model to predict the change of HR expends for some enterprise customers within the next 5 years. In the whole modeling process, we give full consideration to validity, feasibility and cost-efficiency of our model.

2 Discussion of the feasibility of the vacancy chain

2.1 Introduction

Although, theoretically, vacancy chain can improve the benefit of people in that chain and hence improve the overall well-being of the whole enterprise and society, whether such group phenomenon can be applicable to every all

industrials or not is open to doubt. That is to say, the question that whether vacancy chain can exert its function as resource allocation on every industries should be answered. So, in this section, we should discuss the feasibility of vacancy chain at the frist.

2.2 Background

Individuals affected by vacancy chains starting in an organization usually are upwardly mobile in small to moderate jumps in status or laterally mobile within their same status; they rarely experience downward mobility.

In systems in which they operate, vacancy chains have a strong influence upon organizational demographics. Such factors as length of service distributions for individuals within the organization, rates of advancement within the system, the age distribution of individuals within particular strata, cohort effects for mobility within the system, and waiting times for advancement.

In the previously work of Guido Fioretti[1] have investigate the effect of vacancy chain under certain circumstances, especially the hierarchies structure of enterprises. We will use his theory to evaluate the feasibility of vacancy chain in certain industries.

2.3 The influence of organizational hierarchies on vacancy chain

2.3.1 Model of organization form

At the beginning, since vacancy chain is a social phenomenon, the spread of information is an indispensable factor in the effect of vacancy chain. Besides, hierarchies level and structure can affect information spread and hence affect the vacancy chainwe introduce some mathematics models which can be used to simulate some kinds of hierarchy structure and distribution of strata of organization in enterprises.

There are mainly four mathematics model we want to introduce.

The first model we introduce is a decreasing exponential distribution from low to high strata. The functional form is

$$f(k) = e^{-\alpha k} \quad (1)$$

where $s \in N$ is the number of strata and $k = 0, 1, 2, 3 \dots s$. This distribution corresponds to the kind of a hierarchy structure with far less managerial positions than subordinate positions. In other words, such structures look like pyramids.

The second model we introduce is a decreasing exponential distribution from high to low strata. The functional form is

$$f(k) = e^{-\alpha(s-k)} \quad (2)$$

where $s \in N$ is the number of strata and $k = 0, 1, 2, 3 \dots s$. This distribution corresponds to an inversed hierarchy with much more managers than subordinates.

Some people may argue that such hierarchy is obviously unlikely in real world. Indeed, we do not want to deny the fact that such structure is unrealistic. However, given that we want to evaluate the effect of different hierarchies on vacancy chain comprehensively, we still introduce this model in this paper. The third

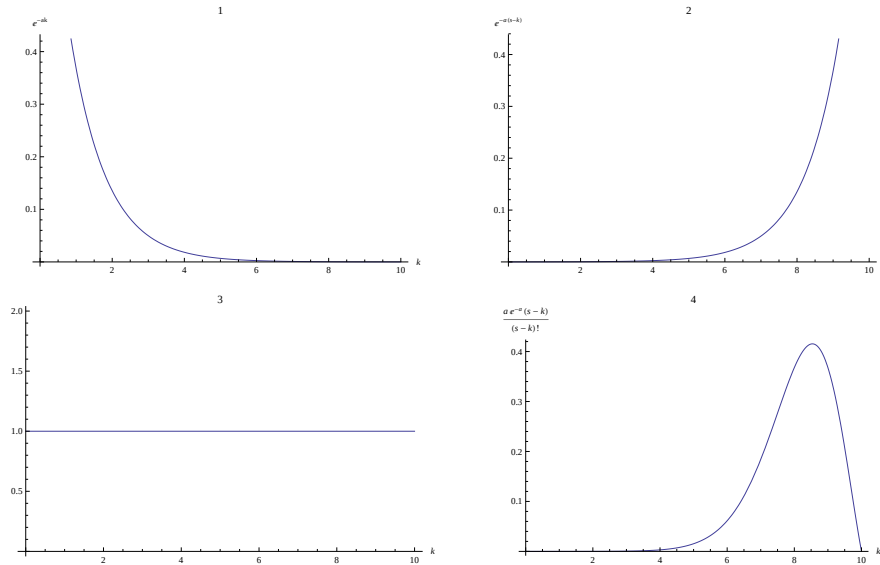


Figure 1: *strata – distribution*

model we introduce is a uniform distribution. The functional form is

$$f(k) = 1 \quad (3)$$

where $s \in \mathbb{N}$ is the number of strata and $k = 0, 1, 2, 3, \dots, s$. The distribution corresponds to an organization where there are as many managers as subordinates. This model is a midpoint between model 1 and model 2.

The fourth model we introduce is a mixed one. That is to say, half the positions assigned to strata according to a decreasing exponential distribution from low to high strata as in model 1. Half the positions assigned to strata according to a positions distribution from high to low strata as in model 2. The functional form is

$$f(k) = e^{-\alpha} \alpha (s - k) / (s - k)! \quad (4)$$

where $s \in \mathbb{N}$ is the number of strata and $k = 0, 1, 2, 3, \dots, s$. This hybrid distribution generates a hierarchy with proportionately less middle managers than a simple pyramid as in model 1.

2.3.2 Analysis

According to work of Guido Fioretti[1], in both of the model 1 and model 2, the spread of information has positive impact on the proportion of vacancy

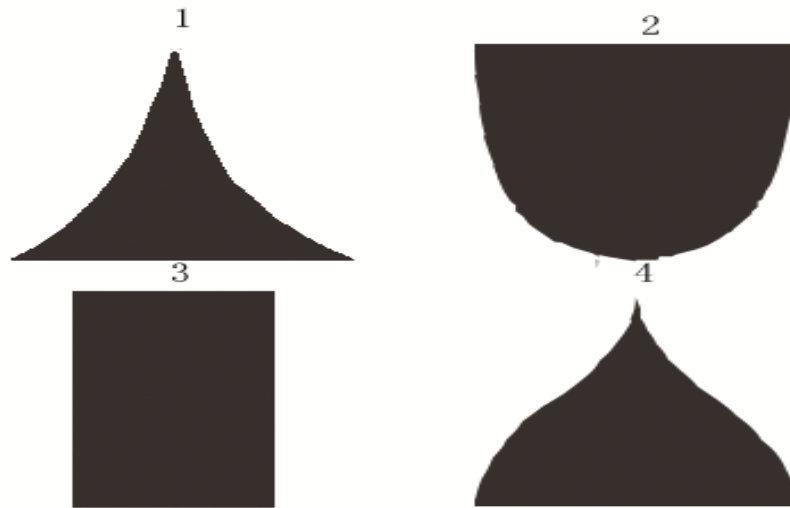


Figure 2: Left to right, the charts obtained when *strata – distribution* is equal to 1,2,3 and 4,respectively.

chains when contiguous strata have a very different number of positions. In model 3, more information never has a positive effect on the proportion of vacancy chains. The impact is even negative. Finally, in model 4, such hierarchy can not only have positive impact on the proportional of vacancy chains when information spreads, but also can make a large use of vacancy chains in order to allocate resources.

2.3.3 Conclusions

To sum up, from the discussion above, hierarchical organizational forms with less middle managers than straight pyramids are most likely to be regulated by vacancy chains. That is to say, vacancy chains can exert its power on industries whose organizational form is similar to model 4.

3 A Vacancy chain model

3.1 Introduction

When studying the behavior of hermit crabs, scientists find that instead of gathering in a random arrangement, hermit crabs tend to behave more rationally. That is to say, hermit crabs will gather around the new shell and form a kind of queue from largest to smallest in size so that every crab can receive benefits.

In order to function as a tool of resources allocation, vacancy chain should not be work automatically. Instead some rules or strategies need to be involved so that everyone's interests can be ensured and enhanced, just like hermit crabs do. Hermit crabs will arrange themselves according to their size, in this section we will mimic such behavior. We will introduce a mathematics model to evaluate employees' competence and then every employee will get a overall score. Then according to these scores, we will use a greedy algorithm to allocate jobs among these employees.

3.2 Assumption

- We assume that the organizational form is similar to the model 4 in former section.
- We assume that job requirement, employees' competence and welfare the job brings can all be quantized.
- We assume that people in relatively lower strata will obey the arrangement conducted by higher strata.
- We assume that the initial arrangement is reasonable. That is to say, job and people in this job are compatible. People with high competence will have high position in enterprise which asks for a high requirement and receive high welfare.

3.3 Procedure

3.3.1 Analysis

At the beginning, we will discuss how to evaluate a person's "size"(competence) and factors that can reflect one's competence.

In 1973 American psychological theorist David Clarence McClelland proposed the term "Competence Model"[2]. He defined competence as five parts, knowledge, skill, self-concept, traits, motives. Figure 3 shows the five parts of competence.

Although the concept of competence model is appealing, how to apply such abstract model to real world is a rather complicated and harsh mission. Indeed,

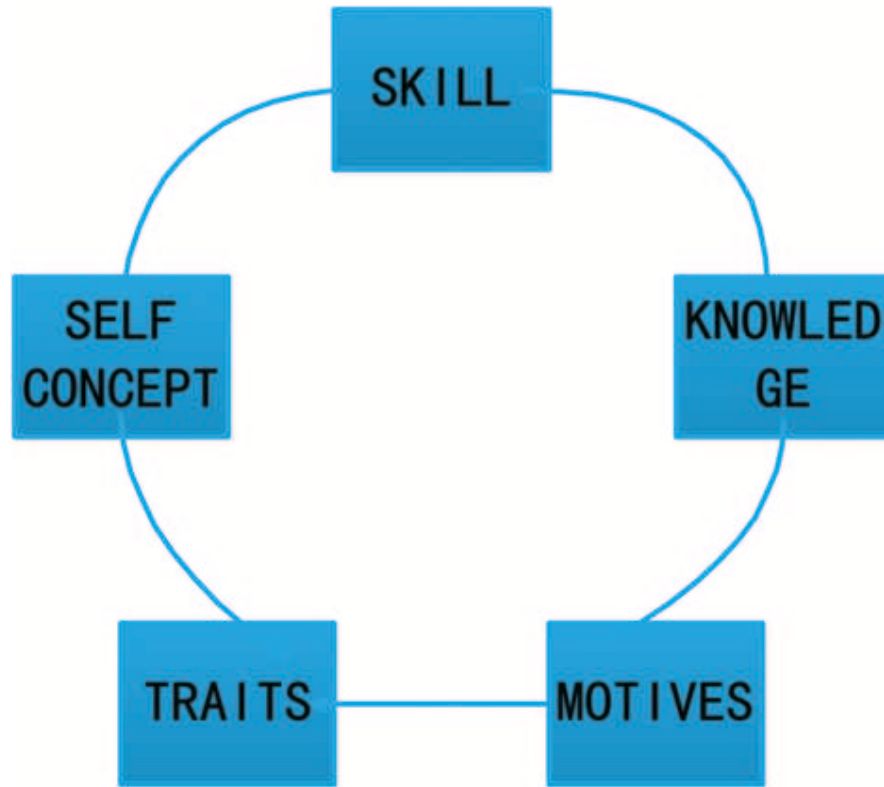


Figure 3: Five parts of competence defined by McClelland

Iceberg Theory is applicable to this case[3]. As stated in figure 4, surface knowledge and skill competences were relatively easy to develop and assess; Core motive and trait competencies at the base of the personality iceberg are more difficult to assess and develop; Self-concept competencies lie somewhere in between.

3.3.2 Model Solution

Furthermore, they experts have developed methods to evaluate one's competencies. According to the work of Lyle M.Spencer and Signe M.Spencer[4], there are mainly eight kinds of method we can use to evaluate employees. Furthermore they proposed a quantitative way to evaluate the correlation coefficients among these eight kinds of method. Table 1 shows Spencer's work.

Since Spencer's work is comprehensive and robust, we will use their conclusion as basis for our model.

In here, we propose a linear equation. This equation is shown below.

$$F(x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8) = a_1x_1 + a_2x_2 + a_3x_3 + a_4x_4 + a_5x_5 + a_6x_6 + a_7x_7 + a_8x_8 \quad (5)$$

$F(x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8)$ is defined as the overall scores for employees. The higher the score is, the more competent the employee is. $x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8$

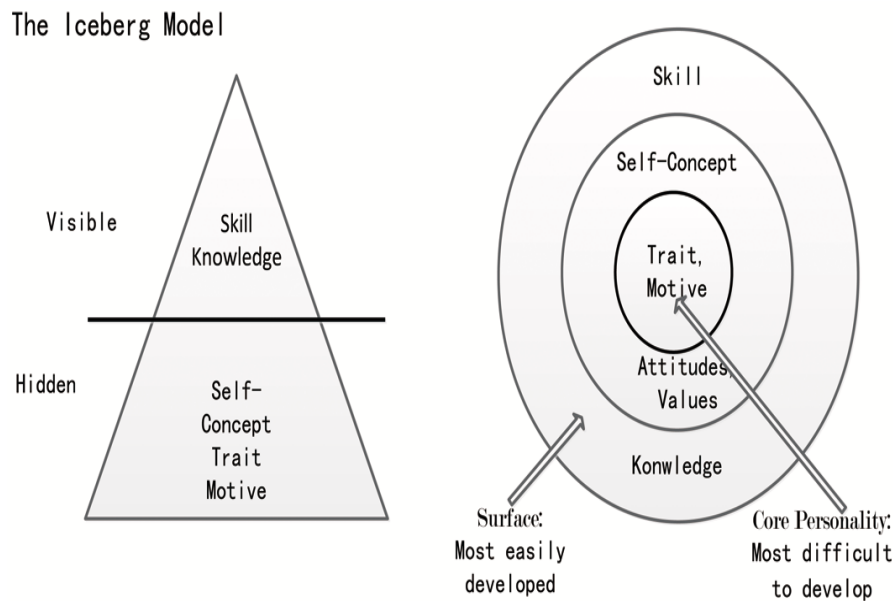


Figure 4: The Iceberg Model and Central and Surface Competences
Source: Spencer and Spencer(1993:11)

evaluation method	R (correlation coefficient)
Assessment center	0.65
Behavioral event interview	0.48-0.61
Work sample test	0.54
Ability test	0.53
Personality test	0.39
Resume	0.38
Recommendation	0.23
Non-behavioral interview	0.05-0.19
source: Spencer, Competence at work	

are the scores of test from Assessment center, Behavioral event interview, Work sample test, Ability test, Personality test, Resume, Recommendation, Non-behavioral interview, respectively. $a_1, a_2, a_3, a_4, a_5, a_6, a_7, a_8$ are the normalization result of those correlation coefficients shown in the table 2. Table 2 shows the actual number of a_1 to a_8 .

3.3.3 An algorithm to allocate jobs in vacancy chain

Now that we develop a method to evaluate peoples competencies. It's time to develop a way to arrange employees just like hermit crabs when vacancy chain appears. We use Greedy Algorithm to allocate jobs. The algorithm are stated as follows:

Before the algorithm, we should use (5) to calculate all employees' overall

evaluation method	$w(\text{weight})$
Assessment center	0.19
Behavioral event interview	0.16
Work sample test	0.16
Ability test	0.16
Personality test	0.12
Resume	0.11
Recommendation	0.07
Non-behavioral interview	0.04

scores.

Once an employee leaves his/her job, according to his/her overall scores, we can limit the employees who will be involved in vacancy chain in those people whose overall scores are lower than the leaving one. This is because individuals affected by vacancy chains starting in an organization usually are upwardly mobile in small to moderate jumps in status or laterally mobile within their same status; they rarely experience downward mobility.

Once a new job appears in a company, then we can compare this job with other jobs that are already existing to estimate its relative position among all the jobs in this company. Remember the assumption that the general welfare of a job is proportional to its requirement and therefore it is also proportional to the overall scores we proposed. And now, we can use this estimated value to limit the employees who will be involved in vacancy chain in those people whose overall scores are lower than the estimated value.

Now that we have defined the domain, we can describe the allocation algorithm.

- According to the scores of eight tests mentioned above, use (5) to calculate the overall scores for every employee in that vacancy chain.
- Sort employees by overall scores in descending order.
- The one who has highest scores will occupy the vacant job; The one who has second highest scores will occupy the job that the highest one leaves, the rest may be deduced by analogy.
- If all employees in chain change their algorithm will end.

3.4 Conclusions

To sum up, from the discussion above, we develop a mathematics model to comprehensively evaluate and calculate employee's competencies. Then by using Greedy Algorithm, we optimize job allocation in vacancy chain so that everyone in this chain can improve their benefits. Just like Adam Smith once said,

the improvement of personal welfare can lead to the improvement of social welfare. Using our model, vacancy chain can improve the interests of individuals, then the interests of enterprises, and finally well-being of the society as a whole.

4 Practice of model

4.1 Introduction

So far, our discussion and model are based on theoretical and ideal situation. We do not consider external factors like competitors in job markets and costs from vacancy chain.

Since the purpose of enterprises is maximize the profits with least costs. In this section, we will consider the limitation of the number of people in vacancy chain to ensure that the cost may not outweigh the benefits it brings. And then, we will run our model to predict the expenses in exchanging jobs.

4.2 Limitation of the number of people in vacancy chain

Every coin has two sides. In other words, vacancy chain can also bring potential negative effects like some overhead costs.

According to the work of Wayne F.Cascio[5], there are mainly three kinds of costs when exchanging job. They are separation cost, replacement costs and training costs. Usually, separation cost consists of exit processing, separation pay and unemployment tax impact and so on. Replacement cost consists of communication of vacancy, pre-employment admin, selection interviews, relocation expense and so on. Training cost consists of formal training, OJT observation, training wages and so on.

Now that we have a brief outlook of costs, it is reasonable to assume that if there are 100 people in vacancy change, that is to say, there 100 people need to exchange their jobs, the costs of changing job maybe extremely high. In this case, perhaps hiring some people from job markets may cost much lower. Indeed, in reality world, employers usually tend to hire people from job markets in order to save money if exchanging cost is much more higher.

In here, we propose a inequality to describe the case mentioned above.

$$(A_1 + A_2 + \dots + A_i) + (B_1 + B_2 + \dots + B_j) \leq C_1 + C_2 + \dots + C_k \quad (6)$$

The inequality shown above can help us answer the question that how many job applicants do need to satisfy both employers and employees. A_n (n for 1 to i) means the costs of exchanging job for employee n. B_n (n for 1 to j) means the costs of hiring person n from outside markets when vacancy chain happens. C_n (n for 1 to k) means the cost of hiring person n from outside markets when vacancy chain does not happen. Usually, the subscript k is equal to subscript j.

This inequality tells us the cost of vacancy chain(cost of exchanging people and hiring some people) should not outweigh the cost of directly hiring people. Some people may argue that both cases we need to hire the same number of people, then why should we use vacancy change. This is easy to explain, although the number of people hired from outside markets is equal, their cost is not. Through vacancy chain we should only hire people for jobs whose requirements are relatively lower and thus the costs are also relatively lower. Let me use an example illustrate it, when selecting a president for a country, the cost is usually extremely high. It's been a practice that once running for a leading position in government, those candidates will compete fiercely on screens and in newspaper for more supporters. Costs are the last thing that is taken into account, for they are all aware the important role an appealing image plays in their campaign. Compared with selecting president, the cost of selecting local governor is much more lower. Through vacancy chain, high positions are occupied by employees in internal enterprise and low positions are left to be occupied by hiring outside people. In this case, it is possible that vacancy chain can save money than directly hiring people from markets.

From the discussion above, we can draw the conclusion that if the inequality can be satisfied then both employees and employers can benefit. Otherwise we should hire people from outside markets instead of using vacancy chain to exchange jobs.

4.3 Model Testing

Now, we will run our model to find out the expenses.

At the beginning, there are four people in an enterprise registered in our website. (see figure 5)

Then employee A left. If directly hiring a person for A' original position costs 10,000 dollars. The costs of exchanging jobs for B, C, D are 3,000, 2,000, 1,000 dollars, respectively and the cost of hiring people for D's position cost 2,000 dollars. Then according to the inequality we mentioned above, we will use vacancy chain instead of directly hiring people. figure 6 shows the change(We hire E to occupy D's original position).

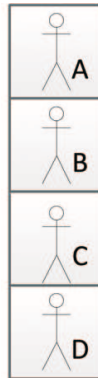


Figure 5:



Figure 6:

Then, employer C left(see figure 7).If directly hiring a person for C's position costs 8, 000 dollars. The costs of exchanging jobs for D, E are 2, 500, 1, 500 dollars, respectively and the cost of hiring people for E's position cost 2, 000 dollars. Then according to the inequality we mentioned above, we will use vacancy chain instead of directly hiring people. Figure 8 shows the change(We hire F to occupy E's original position).



Figure 7:

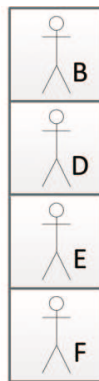


Figure 8:

Then, a new job is created above B's position(see figure 9).If directly hiring a person for this new position costs 11, 000 dollars. The costs of exchanging jobs for B, D, E, F are 4, 000, 3, 000, 2, 000, 1, 000 dollars, respectively and the cost of hiring people for F's position cost 3, 000 dollars. Then according to the inequality we mentioned above, we will directly hire people for the new position instead of using vacancy chain to exchange job. Figure 10 shows the change(We hire H to occupy the new position).



Figure 9:



Figure 10:

If in the next five years, there will be m hirings and n exchanges and the average costs are x dollars and y dollars respectively. Then, we predict the change of HR expends for some enterprise customers within these five years is $x * m + y * n$.

4.4 Conclusions

So far, we have discussed the conditions need to meet so that both employees and employers can benefit when vacancy chain happens. Besides, we run our model to predict the cost of change of HR.

5 Strengths and Weaknesses

5.1 Strengths

Our competency evaluation model is based on quantitative analysis, so our results of evaluation are objective and efficient.

The conditions need to meet so that vacancy chain can function well are based on expert's research, so our results are more convincing and persuasive.

Our prediction is also based on quantitative analysis, enterprises can use this quantitative results to adjust their arrangement.

5.2 Weaknesses

Our model(in section 3)is restricted by limited data.

We define several assumptions and restrictions to simplify our model building process, which may undermine its feasibility in real world.

In the process of prediction(in section 4), we don't get the exact value of the change costs due to lack of data.

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6 Advertisement

A green soldier figurine in a box with text overlays. The box is white with a red interior and a decorative floral pattern on the exterior. The soldier is holding a rifle and is positioned in front of a circular mirror. The background is a warm, textured wall with a potted plant on the left and a framed picture on the right. A string of pearls lies on the surface in front of the box. The text "Unsatisfied with your job position?" and "Need to change a job?" is written in red cursive at the top left. The text "We provide a robust evaluation standard to test your abilities." and "We develop an optimal method to help you find the best position in enterprises." is written in green at the bottom left. A large, white, diagonal watermark "COME TO JOIN US" is overlaid across the center of the image.

Unsatisfied with your job position?
Need to change a job?

COME TO JOIN US

We provide a robust evaluation standard
to test your abilities.
We develop an optimal method
to help you find the best position in enterprises.

